



TEMASEK JUNIOR COLLEGE

Preliminary Examinations 2015

BIOLOGY

Higher 2

Paper 2 Core Paper

9648/02

Friday, 28 August

2 hours

Candidate Name: _____

Civics Group: _____

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Part I, Part II and Part III)

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten the answer papers securely and hand up separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/8
3	/13
4	/9
5	/10
6	/10
7	/ 12
8	/ 8
Section B	
9 / 10	/20
TOTAL	/100

This document consists of **22** printed pages and **1** blank page.

[Turn over

1 Fig. 1.1 shows two organelles from an animal cell involved in digestion.

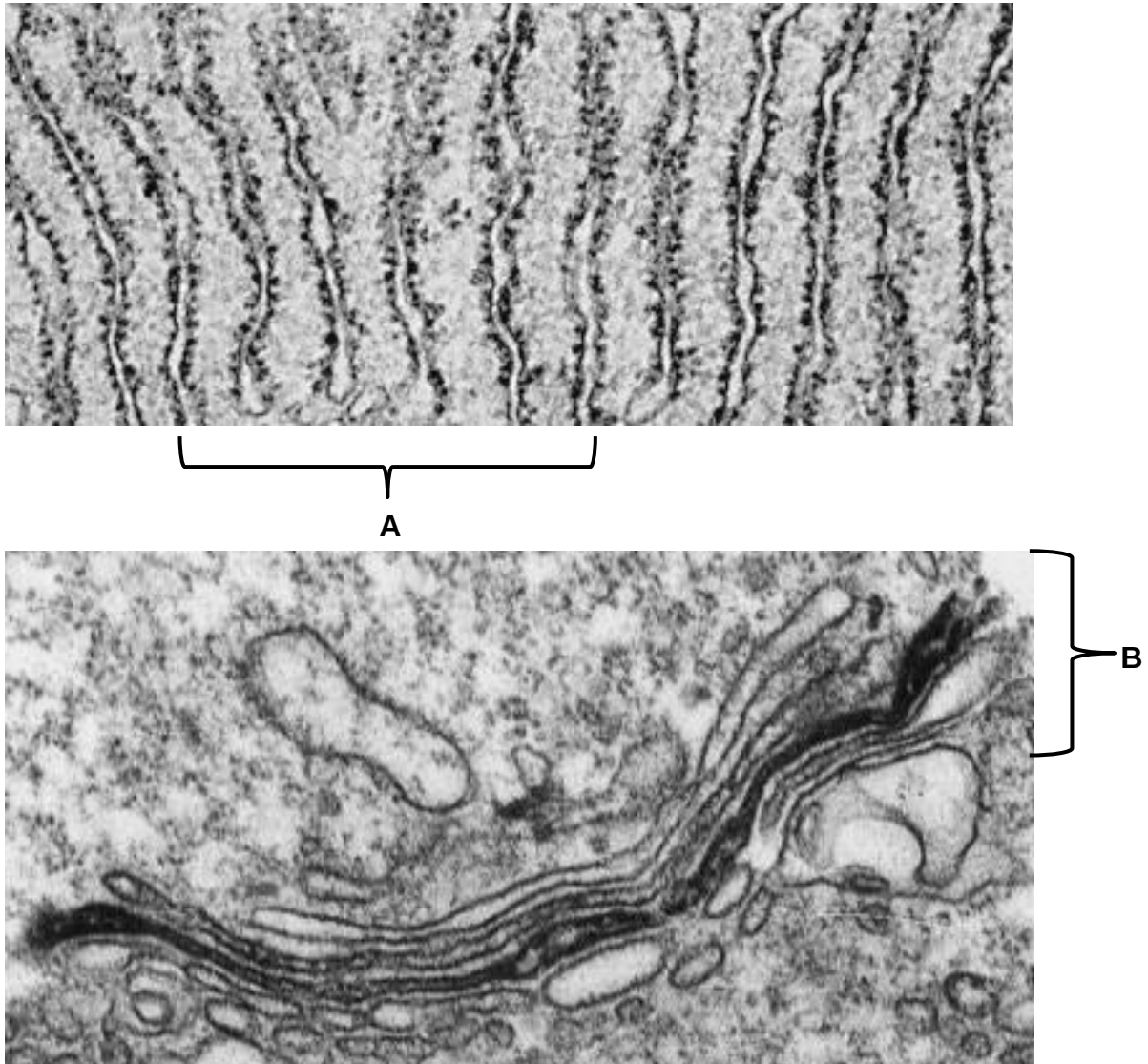


Fig. 1.1

(a) (i) Identify organelles A and B.

A: _____

B: _____ [2]

(ii) Describe how a biomolecule synthesized in **A** is transported to **B**.

[2]

(b) The passage of most molecules through the cell membrane is regulated by proteins. State the roles of these proteins.

[2]

A study was conducted to investigate the effect of disrupting the plasma membrane of a cell. The plasma membrane was disrupted by poking it with a fine needle.

Fig. 1.2 shows the observation made by the researcher 10 s after the membrane was disrupted.

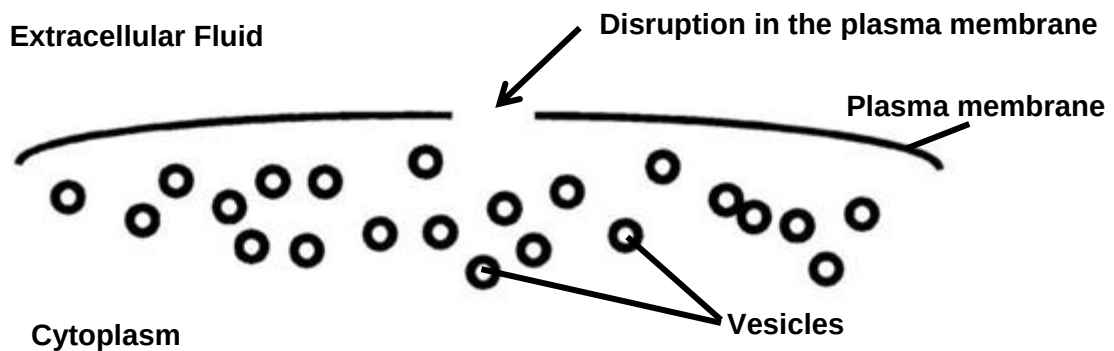


Fig. 1.2

After which, the plasma membrane was resealed, as shown in Fig. 1.3.

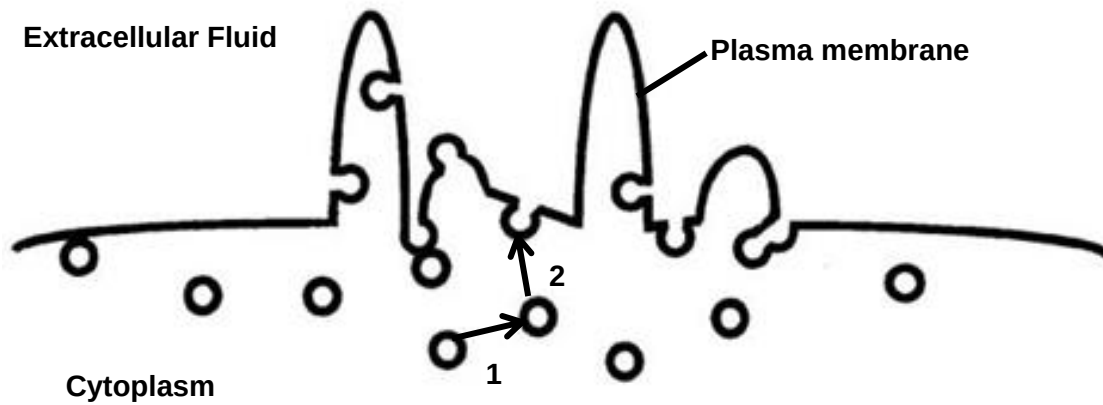


Fig. 1.3

- (c) Numbers **1** and **2** in Fig. 1.3 represent a sequence of events occurring in the cell. Describe the process that is taking place in Fig. 1.3.

[4]

[Total: 10]

- 2 The double helix structure of DNA discovered by Watson & Crick in 1953 is shown in Fig. 2.1.



Fig. 2.1

- (a) Explain how the stability of the DNA structure is maintained.

[3]

RNA molecules that possess catalytic activity are known as ribozymes. One well-known ribozyme is found in the ribosome. The catalytic site of ribosomes, which holds the growing peptide chain and aminoacyl-tRNA in place, is made of RNA.

- (b) (i)** State the name of the enzyme that is found in ribosomes and describe the reaction it catalyses.

[2]

- (ii)** Suggest how the RNA molecules form the catalytic site of the ribosome.

[2]

Lactimidomycin, a potential anti-cancer drug, is found to selectively inhibit translation. Laboratory studies involving lactimidomycin have resulted in an accumulation of dipeptides in the cells.

- (c)** Suggest how lactimidomycin works to inhibit translation.

[1]

[Total: 8]

- 3 In a non-dividing cell (as shown in Fig. 3.1), chromatin can be observed in two functional states – euchromatin or heterochromatin.

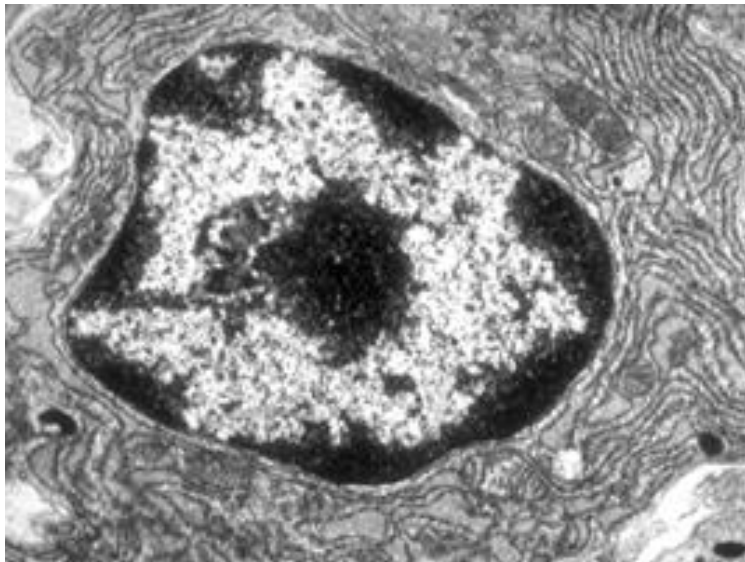


Fig. 3.1

- (a) Explain the difference between the appearance of euchromatin and heterochromatin.

[2]

Gene expression can be controlled via chromatin modifications. This is a form of epigenetics, which is defined as heritable changes in gene activity and expression that occur without alteration in DNA sequence.

- (b) (i) State **two** forms of epigenetic modifications that can prevent gene expression in eukaryotes.

[2]

(ii) Outline how the **two** forms of epigenetic modifications stated in (b)(i) can prevent transcription.

[4]

The Chromatin Immunoprecipitation (ChIP) Assay is a popular technique used to characterize chromatin modifications and analyze the occupancy of transcription factors at specific loci or throughout the genome of a cell.

(c) Explain the role of transcription factors in the control of gene expression.

[3]

(d) Suggest and explain the likely observation which will be made on the occupancy of transcription factors on certain genes in cancerous cells.

[2]

[Total: 13]

✂ End of Section A [Part I] ✂

[Turn over



TEMASEK JUNIOR COLLEGE
H2 BIOLOGY
JC2 / IP YEAR 6 2015

PRELIMINARY EXAMINATIONS

BIOLOGY SECTION A [PART II]:

Structured Questions

Name: _____ ()

Civics Group: _____/14

Q4	/9
Q5	/10
Q6	/10

- 4 There is growing evidence that lifestyle factors may affect the health and lifespan of an individual by affecting telomere length. Factors such as smoking, exposure to pollution, lack of physical activity, obesity, stress and unhealthy diet increase the rate of telomere shortening.

Fig. 4.1 shows the structure of the telomeric DNA.



Fig. 4.1

- (a) Describe how the structure of telomeric DNA helps telomeres carry out its function.

[3]

- (b)** State and explain the significance of an increase in the rate of telomere shortening.

[2]

Shorter telomeres can also induce genomic instability, which increases the rate of mutation and may contribute to the development of cancer. It is observed that most cancer cells have elevated telomerase activity and shorter telomere length, relative to normal cells.

- (c)** Outline how genomic instability and elevated telomerase activity can contribute to the development of cancer.

[4]

[Total: 9]

5 Fig. 5.1 shows an electron micrograph of bacteriophages.

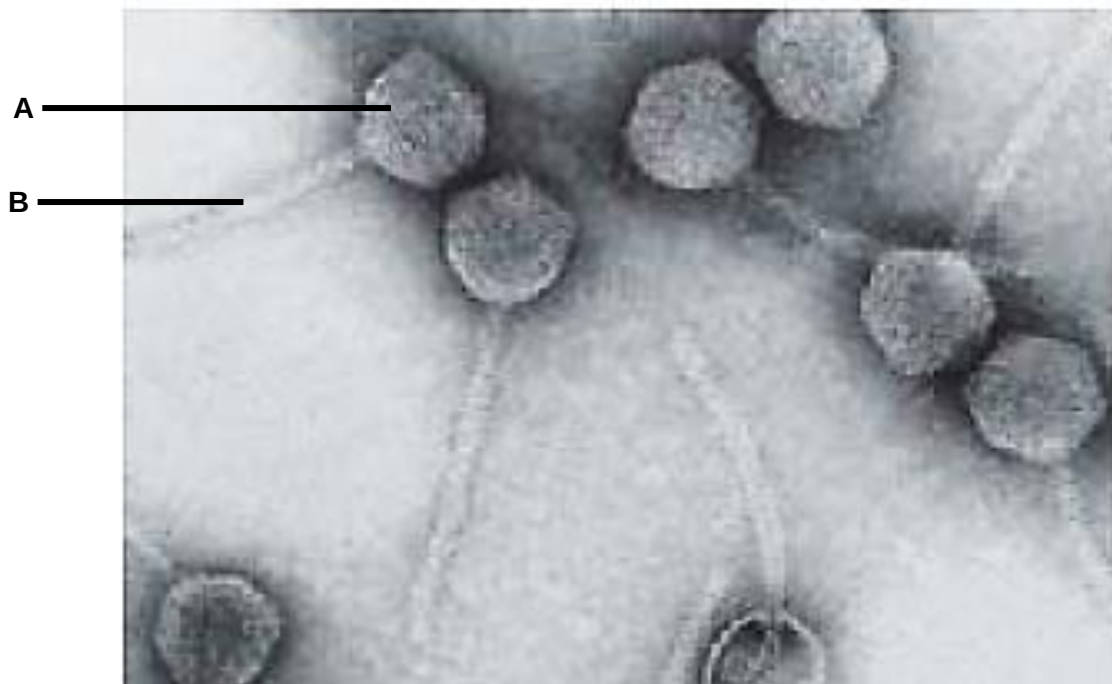


Fig. 5.1

(a) Identify structures **A** and **B**.

Structure **A**: _____

Structure **B**: _____

[2]

(b) Using a named example, describe the processes that lead to the formation of prophage.

[3]

A certain type of bacteriophage inserts its genome exactly between the *lacZ* and *lacY* genes of the bacterial *lac* operon, as shown in Fig. 5.2.

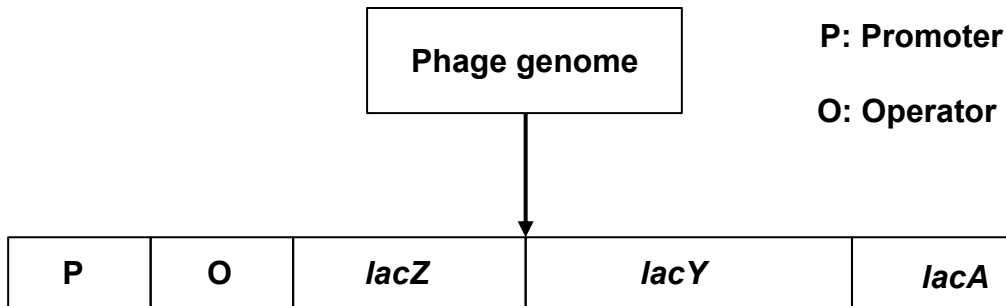


Fig. 5.2

The bacteriophages were added to *E. coli* grown in a nutrient medium that lacks lactose and glucose. Fifteen minutes following infection, lactose was added to the medium to induce the formation of new bacteriophages. Fig. 5.3 shows the effect of lactose on the number of the bacteriophages in the medium.

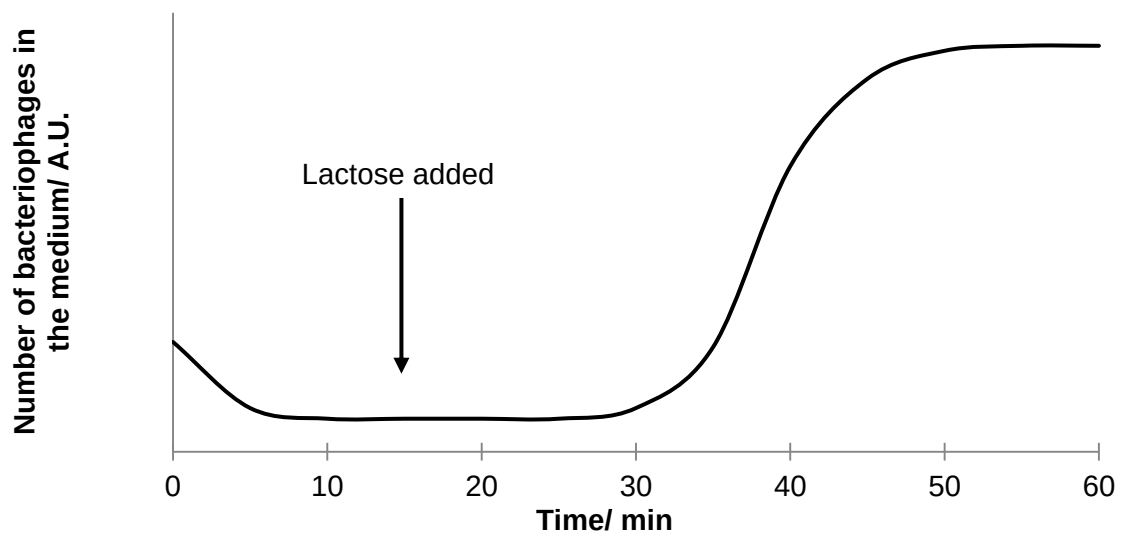


Fig. 5.3

(c) (i) Describe the effect of lactose on the *lac* operon.

[3]

- (ii) Explain why the number of bacteriophages in the medium increased after the addition of lactose.

[2]

[Total: 10]

- 6 When two pure-breeding albino freshwater snails *Physa heterostropha* were crossed, all the first generation progeny were pigmented. The first generation snails were then crossed, which results in a second generation consisting of 740 pigmented snails and 562 albino snails. The control of pigmentation is an example of epistasis resulting in a ratio that is close to 9:7.

(a) State the meaning of the term *epistasis* in this context.

[3]

(b) Use the symbols, **A**, **a** and **B**, **b** to draw a genetic diagram to explain the results shown in the second generation.

[4]

A chi-squared test was carried out on the results of the second cross.

Phenotype	Observed number (O)	Expected number (E)	$\frac{(O-E)^2}{E}$
Pigmented	740	732	0.087
Albino	561	569	0.112
χ^2 value = 0.20			

Part of the critical values of the chi-squared distribution is shown below.

degrees of freedom	probability						
	0.995	0.975	0.9	0.5	0.1	0.05	0.01
1	.000	.000	.016	0.455	2.706	3.841	6.635

- (c) Explain how the chi-squared calculated value of 0.20 supports the statement that epistasis is the correct explanation of these results.

[3]

[Total: 10]

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PRELIMINARY EXAMINATIONS

BIOLOGY SECTION A [PART III]:

Structured Questions

Name: _____ ()

Civics Group: _____/14

Q7	/12
Q8	/8

- 7 There are many different species of the *Drosophila* fruitfly. Three of these species, *Drosophila pseudoobscura*, *D. persimilis* and *D. miranda*, are thought to be closely related.

Samples of these three species were collected from the western United States of America. The base sequences of four regions of DNA of each species were sequenced.

The divergence of these base sequences in *D. pseudoobscura* and *D. persimilis* from the sequences in *D. miranda* was calculated. The results are shown in Table 7.1.

Table 7.1

DNA region	<i>Drosophila</i> species	Percentage divergence of base sequence from that of <i>D. miranda</i> %
Non-coding region 1	<i>pseudoobscura</i>	2.8
	<i>persimilis</i>	2.4
Non-coding region 2	<i>pseudoobscura</i>	8.1
	<i>persimilis</i>	7.3
Coding region 1	<i>pseudoobscura</i>	2.1
	<i>persimilis</i>	2.0
Coding region 2	<i>pseudoobscura</i>	1.9
	<i>persimilis</i>	1.7

[Turn over

- (a) With reference to Table 7.1, using the evidence from the non-coding regions, explain why *D. miranda* may be more closely related to *D. persimilis* than to *D. pseudoobscura*.

[3]

Other than the four DNA regions that were used in this study, a high proportion of non-coding DNA sequences were often used for further evolutionary studies.

- (b) Explain why non-coding DNA is suitable for constructing phylogenetic relationships.

[3]

- (c) State the importance of variation in the coding region for evolution to occur.

[2]

The area where *D. pseudoobscura* is found is separated from the areas where the other two species are found by a high range of mountains.

Fig. 7.1 shows where these species naturally occur.

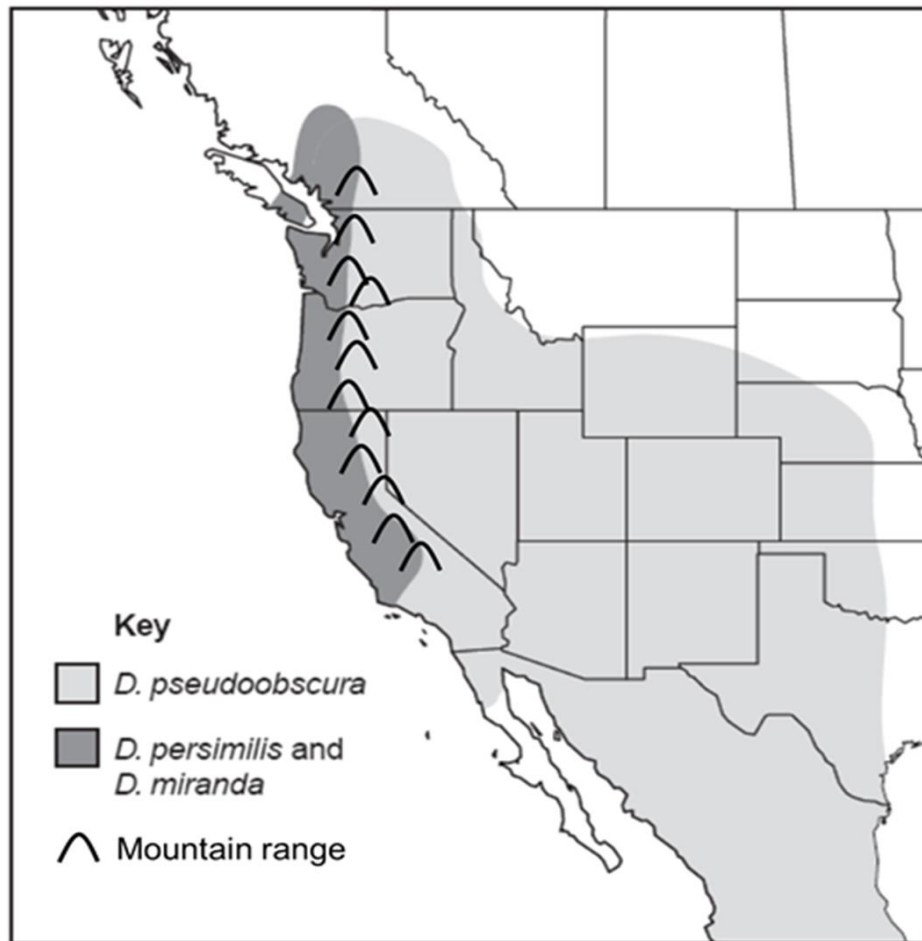


Fig. 7.1

- (d) Outline how the species *D. pseudoobscura* could have evolved from a population of *D. miranda*.

[4]

[Total: 12]

[Turn over

- 8 Destruction of the β -cells in the islets of Langerhans in the pancreas results in a disease condition known as diabetes mellitus type 1.

Table 8.1 shows the changes to blood glucose levels of a non-diabetic individual and a diabetic over the course of a day. The two individuals involved are assumed to be of comparable height and weight, and have similar lifestyles.

At 12 pm, both of them ate the same lunch, and at 3 pm, both engaged in the same series of exercises. At meal times, blood glucose was measured just before the meal.

Table 8.1

Time	Blood glucose level / mg per 100ml of blood	
	Diabetic	Non-diabetic
12 pm	114	90
1 pm	178	92
2 pm	201	91
3 pm	219	91
4 pm	148	90
5 pm	105	90
6 pm	75	91

- (a) With reference to Table 8.1,

(i) state the difference between the blood glucose concentrations of the diabetic and non-diabetic individuals between 3 pm and 5 pm;

_____ [1]

(ii) explain why there is a difference in blood glucose concentrations of the diabetic and non-diabetic individuals between 3 pm and 5 pm.

 _____ [2]

In diabetes mellitus type 2, scientists have hypothesized that insulin is able to bind normally to its receptor, but the signal is not able to elicit the proper cellular response. Fig. 8.1 shows a part of the signaling pathway for insulin.

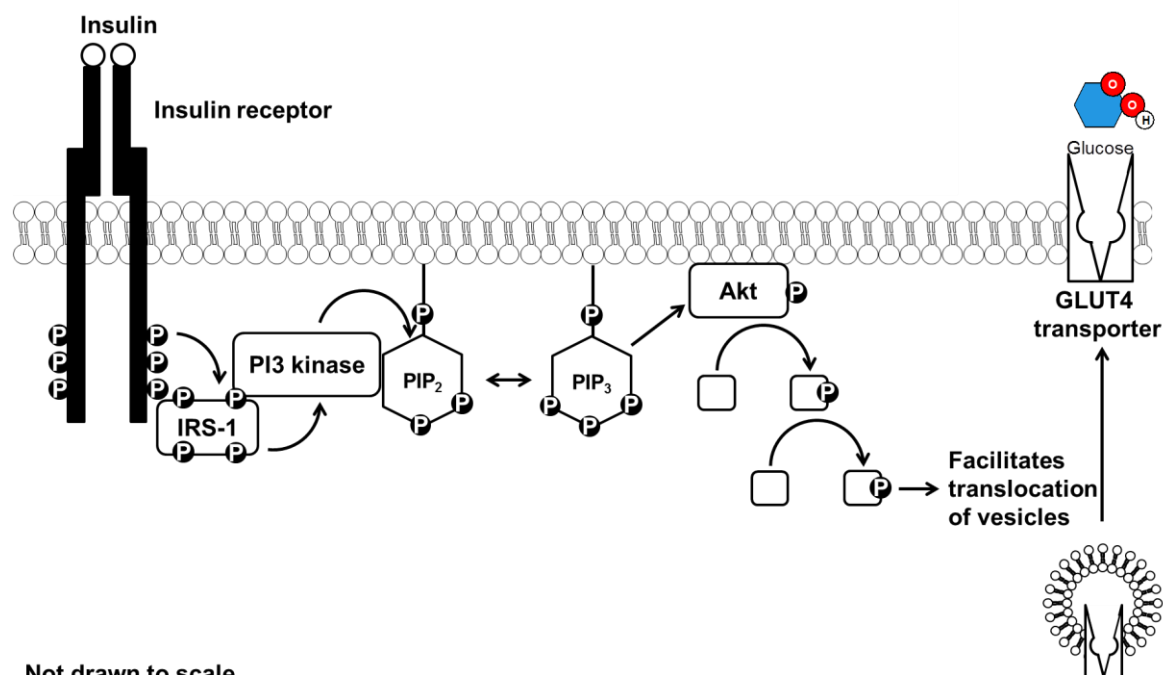


Fig. 8.1

(b) Describe how phosphorylated IRS-1 decreases blood glucose concentration in a non-diabetic.

[3]

(c) Describe **one** significance of having a cell signaling system.

[2]

[Total: 8]

SECTION B**Answer one question.**

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9 (a)** Outline the molecular structure of phospholipids in relation to their function in cell membranes. [6]
- (b)** Discuss the roles of NAD and NADP. [6]
- (c)** Compare oxidative phosphorylation and photophosphorylation. [8]
-
- 10 (a)** Explain how the structure of collagen and haemoglobin are related to their function. [8]
- (b)** Outline the main features of anaerobic respiration. [6]
- (c)** Compare competitive and non-competitive inhibitors. [6]